

Assignment 2

KG 573: Foundations of Database

Part A: 1NF, 2NF, 3NF

Task 1: Consider the following Student table. (1NF)

StudentID	StudentName	CoursesEnrolled	PhoneNumbers
101	Alice Smith	DBMS, OOP, Networks	123-4567, 987-6543
102	Bob Brown	DBMS, AI	555-2222
103	Carol White	OOP, Networks, Security	888-1111, 444-3333

- Identify the problems in this table that violate the rules of First Normal Form (1NF). Explain clearly which aspect of 1NF is being violated and why.
- Normalize the Student table into 1NF. Provide the resulting relation(s) and a short explanation of how the changes improve data organization.

Task 2: Consider the following Enrolment Table. (2NF)

Note: A student can enrol in many courses, and a course has many students.

StudentID	Major	CourseID	CourseTitle	InstructorID	InstructorName	InstructorOffice	Grade
101	CS	DBMS101	Database Systems	I11	Dr. Khan	Room 304	A
101	CS	OOP201	OOP in Java	I12	Prof. Lee	Room 210	B+
102	IT	DBMS101	Database Systems	I11	Dr. Khan	Room 304	A-
103	CS	NET150	Computer Networks	I13	Dr. Silva	Room 118	B
102	IT	NET150	Computer Networks	I13	Dr. Silva	Room 118	B+

- Identify the primary key for the Enrolment table and justify your choice.
- List all the functional dependencies that hold in this table.
- Decompose the table into Second Normal Form (2NF). Provide the resulting relations and briefly explain how the decomposition removes partial dependencies.

Task 3: Consider the following Table. (3NF)

EmpID	EmpName	DeptID	DeptName	DeptPhone
201	Alice Green	D01	Computer Science	555-1111
202	Bob Brown	D02	Mathematics	555-2222
203	Carol White	D01	Computer Science	555-1111
204	David Khan	D03	Physics	555-3333
205	Eva Silva	D02	Mathematics	555-2222

- List the **non-trivial functional dependencies** implied by the notes and the data.

- Decompose the table into Third Normal Form (3NF). Provide the resulting relations and explain why the final set of relations is in 3NF.

Part B: BCNF, 4NF, 5NF

Task 1: Consider the following table. (BCNF)

StudentID	CourseID	InstructorID
101	DBMS101	I11
101	OOP201	I12
102	DBMS101	I11
103	NET150	I13
104	OOP201	I12

- Identify all functional dependencies in the table.
- Determine whether the table is in Boyce-Codd Normal Form (BCNF). If not, decompose it into BCNF relations.
- Show the final set of relations in BCNF and explain why they satisfy the BCNF requirements.

Task 2: Consider the following Table. (4NF)

StudentID	Language	Hobby
101	English	Chess
101	English	Guitar
101	Spanish	Chess
101	Spanish	Guitar
102	French	Painting
102	French	Hiking
102	German	Painting
102	German	Hiking

- Identify the multivalued dependencies (MVDs) that exist in this table.
- Explain why these MVDs violate Fourth Normal Form (4NF).
- Decompose the table into 4NF relations. Provide the final relations and explain how the decomposition resolves the violation.

Task 3: Consider the table. (5NF)

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1
S2	P1	J1

- Identify the join dependency in the Contract relation.
- Explain why the table is not in Fifth Normal Form (5NF).

- Decompose the Contract table into 5NF relations. Provide the final set of relations and explain how they satisfy 5NF.